

Week 6 Software Lesson: Logistic Regression in R

Goals

The goal of this software lesson is to learn how to

- fit a logistic regression model and obtain estimates for the effect of a predictor in terms of log odds and in terms of odds ratios, and
- obtain model diagnostics to check assumptions and evaluate the model fit.

Required Packages

You will need to install the following packages in R for this software lesson:

- jtools
- tidyverse
- ResourceSelection

DATASET

In this software lesson, you will use the Core Temperature dataset. The dataset, data dictionary, and dataset introduction can be found on Canvas, titled *Core-Temp.csv*, *Core-Temp-Data-Dictionary.pdf*, and *Core-Temp-Dataset-Intro.pdf*, respectively.

```
core_temp <- read.csv(file = file.choose(), header = TRUE)
```

LOGISTIC REGRESSION

Thus far in the course, you have focused on *linear* regression, which is used when you have a continuous outcome. But, if the outcome is not continuous, other types of regression methods are necessary. This software lesson builds on what you have learned thus far, but focuses on *logistic* regression, which is used when you have a binary outcome.

In the Core Temperature study, the researchers were interested in examining the relationship between the development of serious infections (**SeriousInfection**) and time weighted average core temperature (**TWATemp**) in Celsius, after adjusting for **Age** at time of surgery (in years), sex (**FEMALE**), body mass index at time of surgery (**BMI**), and Charlson score (**CharlsonScore**). In this example, **TWATemp** is the predictor of interest, and **Age**, sex, **BMI**, and **CharlsonScore** are considered to potential confounders.

Fit logistic regression model

When you fit a logistic regression model, you can obtain regression coefficient results in terms of log odds or odds ratios. The model is fit on the log-odds scale, but the results are often more interpretable when the values are converted to odds ratios.

Estimates in terms of log odds

- To fit a logistic regression model and obtain log odd estimates, use the code you have seen thus far, but change the distribution family of the error term to `family = binomial(link="logit")` for a binary response variable:

```
model1 <- glm(SeriousInfection ~ TWATemp + Age + factor(FEMALE) + BMI + CharlsonScore,
              data = core_temp, family = binomial(link="logit"))

summ(model1, confint = TRUE, ci.width = 0.95, digits = 4)
```

The output is similar to what you have seen previously, except the regression coefficient values under the column `Est.` and the lower and upper limits to the confidence interval are in terms of log odds. For example, the regression coefficient value for `TWATemp` is 0.230, which means that for every one-degree (in Celsius) increase in core temperature, the log odds of a developing serious infection increases by 0.230 (holding all else constant).

Estimates in terms of odds ratios

- To fit a logistic regression model and obtain odds ratio estimates, use the code you have seen thus far, but in the `summ()` function, also include the `exp = TRUE` argument to report exponentiated coefficients (i.e., odds ratios) and confidence intervals:

```
summ(model1, confint = TRUE, ci.width = 0.95, digits = 4, exp = TRUE)
```

The output is similar to what you have seen previously, except the regression coefficient values under the column `Est.` and the lower and upper limits to the confidence interval are in terms of odds ratios. For example, the regression coefficient value for `TWATemp` is 1.2587 (which is $\exp(0.230)$), which means that for every one-degree (in Celsius) increase in core temperature, the odds of a developing serious infection increases by 25.87% (holding all else constant).

Evaluate model diagnostics

As with any statistical method, you need to check model assumptions and evaluate model fit. Some code from previous software lessons also apply for logistic regression:

- `lrtest()` for Likelihood Ratio Test
- `vif()` for checking multicollinearity

Only new model diagnostic topics are presented in this software lesson.

Linearity of continuous predictors

One of the assumptions for logistic regression is **L**inearity, meaning the continuous predictor(s) needs to be linearly related to the *log odds* of the binary outcome. You can check this by examining a scatterplot of the model-predicted log odds (i.e., logit) verses the continuous predictor and look to see if the form is approximately linear.

- To create scatterplot(s) of model-predicted logit vs. continuous predictor(s),

- first, create a new object of a subsetted dataset using the `subset()` function that contains only the continuous predictor(s) that are in the model.
- Then, create a new object that contains the names of the variables (`predictors <- colnames(mydata)`).
- Next, create a new object of the model-predicted probabilities using the `predict()` function, specifying the model to obtain the predicted log-odds (logit) of each observation.
- Last, use the `pivot_longer()` function from the `{tidyverse}` package to restructure the data, piping (`%>%`) into the `ggplot()` function from the `{tidyverse}` package to create a scatterplot of logit vs. continuous predictor, specifying
 - * the `aes()` function to indicate the predictor values for the x-axis (`x = predictor.value`) and logit on the y-axis (`y = logit`),
 - * the `geom_points()` function to make points as the geometric shapes for the plot (also known as scatterplot) and denoting the size (`size =`) and transparency (`alpha =`) of the points,
 - * the `geom_smooth()` function to plot “best-fit” lines/curves (`method = "loess"` for a “best-fit” smooth curve; `method = "lm"` for a “best-fit” linear line) and denoting color (`color = "<color"`) and suppressing the SE bands for each line (`se = FALSE`),
 - * the `facet_wrap()` function to create a multi-panel plot arrangement based on the predictors (`facets = ~ predictors`) and allowing the x-axis scales to vary for each plot (`scales = "free_x"`), and
 - * the `theme_bw()` function to modify the theme of the plot:

```
## Subsetting the dataset by selecting only the columns
## with the model's continuous predictors
mydata <- subset(core_temp, select=c(TWATemp, Age, BMI, CharlsonScore))

## Saving variable names for plotting
predictors <- colnames(mydata)

# Obtaining/saving the model-predicted probabilities for each observation
mydata$logit <- predict(model1)

## Restructuring the dataset to then plotting logit vs. continuous predictor
pivot_longer(mydata, names_to = "predictors", values_to = "predictor.value", -logit) %>%
ggplot(aes(x = predictor.value, y = logit)) +
  geom_point(size = 0.5, alpha = 0.5) +
  geom_smooth(method = "loess", color = "blue", se = FALSE) +
  geom_smooth(method = "lm", color = "red", se = FALSE) +
  facet_wrap(facets = ~ predictors, scales = "free_x") +
  theme_bw()
```

The output produces scatterplots of logit vs. continuous predictor(s), one scatterplot for every continuous predictor in the model. For the Core Temperature study example, there were four continuous predictors in the model, so four scatterplots are created.

The blue curve is a “best-fit” curve to the data, called loess - a smoothing curve - and doesn’t assume any sort of form/model of the data. The red line is a “best-fit” *linear* line to the data, where the model is assumed to be a straight line. To evaluate the the linearity of the continuous predictors, you compare the blue curve to the red line in each of the plots. If the two follow a similar trajectory or are relatively the same, then the linearity assumption is met. If the blue curve has a different form (e.g., curved) than red line, then the linearity assumption isn’t met for that predictor.

Hosmer-Lemeshow Goodness-Of-Fit test

One test that can be used to check for the goodness-of-fit of the logistic regression model is the Hosmer-Lemeshow test. The Hosmer-Lemeshow test provides an overall or global assessment of how well the (logistic) model fits the data. The null hypothesis is that the model fits the data well.

- To conduct the Hosmer-Lemeshow test, use the `hoslem.test()` function from the `{ResourceSelection}` package, specifying the binary outcome variable first (Note: this must be denoted as a 0/1 variable), followed by the expected values (fitted observations from the model) for the observations (e.g., `fitted(model1)`):

```
hoslem.test(core_temp$SeriousInfection, fitted(model1))
```

The outcome displays the test statistic (**X-squared**), the degrees of freedom (**df**), and the **p-value** for the test.

Additional Resources

For additional R Resources on logistic regression, see:

- [Logistic Regression in R](#) (Mark Bounthavong in RPubs)
- [Logistic regression with one continuous explanatory variable](#) (Carpentries)
- [Logistic Regression Assumptions and Diagnostics in R](#) (STHDA)
- [Logistic regression in R - Part 2 \(Goodness of fit\)](#) (from UCLA)